

Myomatlas

Decision support for abnormal uterine bleeding

Case	UMD_221129_003
Modality	MRI T2WI sagittal
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PATIENT CONTEXT

Age	34
Menopausal status	Premenopausal
Bleeding severity	Heavy bleeding
Fertility desire	Fertility desired

SUMMARY

MRI T2WI sagittal shows six leiomyomas (PALM-COEIN 'L'). Two are transmural hybrids reaching both the cavity and the serosa (id 1, large 27.14 ml, provisional 2-5; id 2, small provisional 2-5) and are the most relevant to the patient's heavy bleeding through their endometrial contact. Ids 4 and 5 are subserosal type 5 (mostly intramural, low bleeding relevance). Id 3 is subserosal, most likely pedunculated type 7 (type 6 cannot be excluded from masks). Id 6 is a small intramural type 4 with no cavity or serosa contact. Given premenopausal status and desired fertility, uterus- and fertility-preserving options are prioritized, especially specialist-led myomectomy for the cavity-contacting hybrids, with medical measures (tranexamic acid, hormonal therapy, GnRH bridge) for bleeding control. FIGO type calls on the two hybrids and on id 3 are provisional because they depend on fine 50 percent proxy thresholds or on pedunculation that masks cannot resolve. Non-structural (COEIN) causes and endometrial pathology cannot be judged from these masks and need history, labs, and appropriate endometrial assessment. A clinician makes the final decisions.

MALIGNANCY ASSESSMENT

Not flagged

The patient is 34 and premenopausal with no reported risk factors (no unopposed estrogen, obesity, PCOS, tamoxifen, Lynch syndrome), no reported rapid growth, and no suspicious imaging features noted in the measurements. She does not meet the malignancy_exclusion_rule criteria (postmenopausal bleeding; age 45+ with AUB or younger with risk factors; rapid growth; suspicious features; persistent bleeding despite treatment). This tool does not diagnose or exclude malignancy; a clinician should reassess if bleeding persists despite treatment or if risk factors emerge.

FINDINGS (6)

Myoma 1, FIGO type 2-5

Hybrid (transmural), submucosal component reaching endometrium (2) and subserosal component reaching serosa (5) | Group: Transmural | Confidence: lower (provisional) | Relevance: high
Volume 27.14 mL | Max diameter 51.3 mm | Intramural 53 percent | Contact: cavity, serosa
contacts_cavity true (distance_to_cavity_mm 0.0) AND contacts_serosa true (distance_to_serosa_mm 0.0), so this myoma is transmural, reaching both the endometrium and the serosa. The cavity side: cavity_surface_pct 0.0 with intramural_pct 53.0 suggests it abuts/contacts the cavity with 50 percent or more intramural bulk (submucosal digit 2). The serosal side: serosa_surface_pct 46.9 with intramural_pct 53.0 (≥ 50 percent intramural) suggests a type 5 subserosal component. Both digits depend on the fine 50 percent proxy threshold (intramural_pct 53.0 is very near 50 percent), so the exact hybrid designation is uncertain.

Management options

- Myomectomy planned to preserve the cavity and myometrium, specialist-led (fertility desired)
- Combined or staged myomectomy addressing both components
- Tranexamic acid
- Hormonal therapy for bleeding control
- GnRH analogue or antagonist as a bridge
- Hysterectomy if childbearing is complete and symptoms are refractory

Myoma 2, FIGO type 2-5

Hybrid (transmural), submucosal component reaching endometrium (2) and subserosal component reaching serosa (5) |
Group: Transmural | Confidence: lower (provisional) | Relevance: high
Volume 2.59 mL | Max diameter 23.2 mm | Intramural 70 percent | Contact: cavity, serosa

contacts_cavity true (distance_to_cavity_mm 0.0) AND contacts_serosa true (distance_to_serosa_mm 0.0), so this small myoma is transmural. Cavity side: cavity_surface_pct 1.2 with intramural_pct 70.3 (≥ 50 percent intramural) points to a type 2 submucosal component that contacts the endometrium with most bulk intramural. Serosal side: serosa_surface_pct 28.5 with intramural_pct 70.3 points to a type 5 subserosal component. Both digits rest on the fine 50 percent proxy threshold and are provisional.

Management options

- Myomectomy planned to preserve the cavity and myometrium, specialist-led (fertility desired)
- Combined or staged myomectomy addressing both components
- Tranexamic acid
- Hormonal therapy for bleeding control
- GnRH analogue or antagonist as a bridge
- Hysterectomy if childbearing is complete and symptoms are refractory

Myoma 3, FIGO type 7

Subserosal, pedunculated; type 6 (subserosal, less than 50 percent intramural) cannot be excluded from masks |
Group: Subserosal | Confidence: lower (provisional) | Relevance: low
Volume 0.77 mL | Max diameter 12.9 mm | Intramural 0 percent | Contact: serosa

contacts_serosa true (distance_to_serosa_mm 0.0) and contacts_cavity false (distance_to_cavity_mm 16.2). intramural_pct 0.0 with serosa_surface_pct 100.0 matches a near-zero intramural fraction and very high exterior-facing surface, consistent with type 7. However, a true pedicle cannot be confirmed from voxel masks, so type 6 versus 7 is not resolvable geometrically.

Management options

- Myomectomy preserving the uterus when fertility is desired
- Laparoscopic myomectomy (stalk resection)
- Observation when asymptomatic

Myoma 4, FIGO type 5

Subserosal, 50 percent or more intramural | Group: Subserosal | Confidence: moderate | Relevance: low
Volume 0.57 mL | Max diameter 13.2 mm | Intramural 100 percent | Contact: serosa

contacts_serosa true (distance_to_serosa_mm 0.0) and contacts_cavity false (distance_to_cavity_mm 8.0). intramural_pct 99.8 is well above 50 percent and serosa_surface_pct is only 0.2, so the bulk is intramural while it reaches the serosa, consistent with type 5. The value is far from the 50 percent line, so not near-boundary.

Management options

- Myomectomy when clinically indicated and fertility is desired
- Observation when asymptomatic
- Symptomatic management for pressure symptoms

- Myomectomy (laparoscopic or abdominal) for bulk symptoms
- Hysterectomy if childbearing is complete and symptoms are refractory

Myoma 5, FIGO type 5

Subserosal, 50 percent or more intramural | Group: Subserosal | Confidence: moderate | Relevance: low
Volume 0.34 mL | Max diameter 9.4 mm | Intramural 87 percent | Contact: serosa

contacts_serosa true (distance_to_serosa_mm 0.0) and contacts_cavity false (distance_to_cavity_mm 20.7). intramural_pct 86.7 is clearly above 50 percent with serosa_surface_pct 13.3, so most of the bulk is intramural while touching the serosa, consistent with type 5 and not near the 50 percent boundary.

Management options

- Myomectomy when clinically indicated and fertility is desired
- Observation when asymptomatic
- Symptomatic management for pressure symptoms
- Myomectomy (laparoscopic or abdominal) for bulk symptoms
- Hysterectomy if childbearing is complete and symptoms are refractory

Myoma 6, FIGO type 4

Intramural, entirely within the myometrium | Group: Intramural | Confidence: higher | Relevance: low
Volume 0.15 mL | Max diameter 7.1 mm | Intramural 100 percent | Contact: none

contacts_cavity false (distance_to_cavity_mm 15.5) AND contacts_serosa false (distance_to_serosa_mm 7.2), with intramural_pct 100.0. No contact with either surface is a direct geometric signal for type 4.

Management options

- Observation when asymptomatic
- Tranexamic acid
- Hormonal therapy for bleeding control
- Myomectomy when clinically indicated and fertility is desired
- Myomectomy for bulk or bleeding symptoms
- Hysterectomy if childbearing is complete and symptoms are refractory

PATIENT EXPLANATION

Your MRI scan shows six fibroids (non-cancerous growths in the wall of the womb). Here is what they mean in plain terms:

- Two of them (one large, about the size of a small plum, and one small) reach all the way from the inner lining of the womb to the outer surface. Because they touch the inner lining, these two are the ones most likely to be causing your heavy bleeding.
- Two more sit mostly inside the muscle wall but just reach the outer surface. These usually cause little or no bleeding.
- One small fibroid sits on the outer surface and may be attached by a thin stalk. It is away from the inner lining and unlikely to cause bleeding.
- One very small fibroid is buried entirely inside the muscle wall, touching neither the lining nor the outer surface, and usually causes few symptoms.

Because you would like to keep the option of having children, the focus is on treatments that preserve your womb. Options include medicines to reduce heavy bleeding (such as tranexamic acid or hormone treatments, and sometimes a short course of a hormone-blocking medicine before surgery), and surgery to remove the fibroids while keeping the womb (myomectomy). For the two fibroids that reach the lining, this kind of surgery is best planned by a specialist, since they span the full thickness of the womb wall. Removing the whole

womb (hysterectomy) is only considered if you no longer wish to have children and symptoms do not improve with other treatments.

Some details cannot be judged from these scan images alone. The images show the womb wall, the cavity, the fibroids and one small harmless cyst, but they cannot rule out a polyp, a condition where lining tissue grows into the muscle (adenomyosis), or lining abnormalities. Other causes of heavy bleeding, such as bleeding or hormone problems, need blood tests and a review of your history. Based on your age and situation, there is no specific sign here that points to cancer, but any diagnosis and treatment choice must be made by your doctor, who can arrange further tests if needed. This report is to support that conversation, not to replace it.

IMPORTANT

This report is decision support, not a diagnosis. A clinician makes the final diagnosis and treatment decision.